

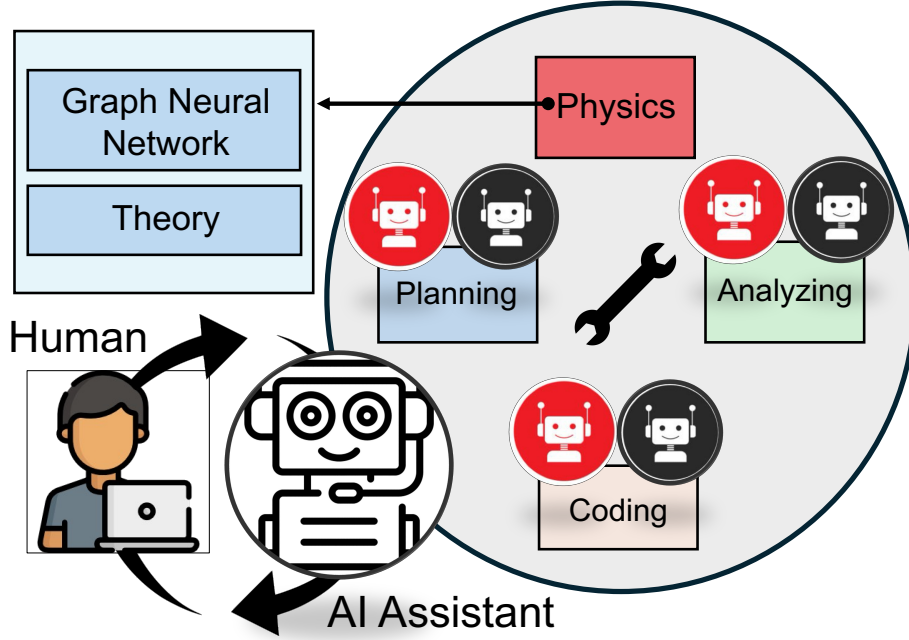


Figure 4: **Overview of the GNN-powered, LLM-based multi-agent system developed here.** The system consists of a human and an AI assistant agent at its core, where the human poses queries, and the AI assistant provides responses, seamlessly steering the problem-solving process with the help of integrated tools. These tools are responsible for various tasks, including planning, coding, and multi-modal analysis, and each incorporates a set of AI agents that dynamically collaborate to solve complex tasks. A key component is the physics tool, which includes newly developed graph neural network (GNN) models to retrieve essential physical parameters (such as the Peierls barrier and potential energy change) as well as physics-based theories (like solute-strengthening theory). The GNN models enable the rapid prediction of fundamental material properties, bypassing the need for costly atomistic simulations. The iterative collaboration between agents within the tools and the seamless interaction between the human and AI assistant allows for efficient resolution of complex materials design challenges.

visualization and plotting. Finally, the Assistant engages the multi-modal agent to conduct a comprehensive analysis of the results.

The advantage of incorporating our newly developed GNN model into the multi-agent system is twofold: (a) it enables the rapid exploration of vast design spaces, providing deeper insights into material behavior across a wide range of compositions, and (b) it significantly accelerates the materials design process by allowing the fast computation of critical input parameters for physics-based theoretical models, such as solute-strengthening theories. To demonstrate the effectiveness of our multi-agent system in these key areas, we conducted two experiments, each detailed below.

Variation of the Peierls barrier in binary and ternary alloys In this experiment, the multi-agent system was tasked with exploring the Peierls barrier across the ternary compositional space, as illustrated in Figure 6(a). Upon receiving the query, the assistant agent activates the planning tool to devise a detailed plan for the task. The embedded planner and reviewer agents collaborate, with the planner drafting the initial plan and the reviewer providing feedback. This feedback loop iterates twice to ensure the accuracy and robustness of the plan. The resulting comprehensive plan outlines the following key steps: (a) Composition Generation: creating a list of all possible binary and ternary compositions in the Nb-Mo-Ta system at 5% intervals (231 total compositions); (b) Computation of Peierls Barrier: using the GNN model to compute the Peierls barrier for all generated compositions; (c) Plotting Results: utilizing the coding tool to plot the computed Peierls barrier values on a ternary diagram representing the Nb-Mo-Ta compositional space; and (d) Data Analysis: applying the analysis tool to examine the plotted data for trends, correlations, and significant insights. The plan also includes a detailed list of functions and input parameters to ensure consistency and precision during execution.